JAYASURYA T APPU

Backend Engineer · Node.js / TypeScript · Payments & Event-Driven Systems
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SUMMARY

Backend engineer with 4+ years in system design and building scalable Node.js/TypeScript APIs and payment systems.

- **Designed the architecture** of an event-driven payout system (RabbitMQ) that pays partners when orders complete.
- Built Node.js REST APIs sustaining **~1,000 requests/sec** for production web and mobile apps.
- Cut API response times by **30%+** through Redis caching and MongoDB query tuning.
- Defined API contracts and **reviewed backend architecture** against Clean Architecture principles.
- Built an analytics and task allocation module that contributed to an **~80% drop** in service cancellations.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, SQL, C
Backend: Node.js, Express.js, FastAPI, REST APIs, gRPC, JWT, Role-Based Access Control (RBAC), FSM workflows
Messaging & Async: RabbitMQ, Redis Streams, background workers, event-driven architecture, Kafka (projects)
Databases: MongoDB (indexing, aggregation, geo queries), MySQL, Redis, ChromaDB
Cloud & DevOps: AWS (EC2), Google Cloud (GCP), Docker, Docker Compose, GitHub Actions (CI/CD), Nginx, Linux, Git
AI / LLM: RAG pipelines, vector embeddings, semantic search, LLM agents, Ollama (local LLMs)
Practices: System design, microservices, API design, Clean Architecture, performance tuning, code reviews, mentoring
Frontend: React.js, Angular

EXPERIENCE

Senior Software Engineer (Backend) — Spericorn Technology Inc, Trivandrum — Aug 2026 – Present

- Own system design, API architecture and production performance for Node.js/TypeScript backend services.

Software Engineer — Carclenx (Pexa), Attingal — Jul 2025 – Aug 2026

Rehired; previously Software Developer here, Jan 2022 – Jun 2022 (Node.js/Express.js APIs, Angular dashboards)

- Built REST APIs serving production web and mobile apps at **~1,000 requests/sec**.
- Built the partner payout flow on RabbitMQ with asynchronous processing, retries and fault-tolerant error handling.
- Built finite state machine (FSM) order workflows integrating payment and messaging providers.
- Cut response times **30%+** with MongoDB indexes, aggregation pipelines, pagination, geo queries and Redis caching.
- Moved long-running jobs out of the request path into Redis Streams pipelines processed by background workers.
- Implemented authentication, session management and role-based access control (RBAC).
- Built worker and staff analytics with task allocation, contributing to an **~80% drop** in service cancellations.
- Automated payroll for **~200 employees**, cutting manual HR effort by **~70%**.

Software Developer — Cordova Educational Solutions, Thiruvananthapuram — Oct 2024 – Jun 2025

- Built Node.js services and REST APIs for an LMS, with auth flows, integrations, and Angular and React.js modules.
- Refactored tightly coupled legacy modules into reusable components, improving performance by **~20%**.
- Shipped features from stakeholder requirements and fixed production issues through root-cause analysis.

Junior Software Engineer & Technical Mentor — Quest Innovative Solutions, Trivandrum — Dec 2022 – Sep 2024

- Built Node.js/Express.js REST APIs and React.js interfaces for client applications, including booking systems.
- Integrated third-party APIs and optimized SQL queries for performance and data consistency.
- Mentored trainees through hands-on projects, technical sessions and code reviews.

PROJECTS

Microservices E-commerce Backend | github.com/Jayasurya-T-Appu/Ecommerce_microservice_docker

- Node.js user, product and order services on MongoDB communicating over gRPC, with JWT authentication, Redis caching and Nginx routing; containerized with Docker and deployed on GCP.

Forge: Build System for C | github.com/Jayasurya-T-Appu/forge-tool-

- Build tool written from scratch in C: scans sources, generates a build plan, compiles with GCC and links; builds itself.

PrivateRAG: Multi-Tenant RAG Platform | github.com/Jayasurya-T-Appu/RAG_AI-AGENT

- Document Q&A on a local LLM (Ollama) using a Node.js API gateway, FastAPI service and ChromaDB, with per-user data isolation through identity propagation and identity-filtered retrieval; runs on Docker Compose.

EC2 CI/CD Pipeline | github.com/Jayasurya-T-Appu/EC2-CI-CD-Pipeline

- Push-to-deploy pipeline rebuilding Docker Compose services on AWS EC2 via a self-hosted GitHub Actions runner.